

Lecture Notes

EC226 Term 1 2025/26

Normality

Readings:

- Wooldridge (2013): Chapter 10

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1 Normality Assumption

Consider the model

$$y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad (1)$$

where all the CLRM assumptions are satisfied, with the exception of CLRM assumption 4, that is, $\varepsilon_i | \mathbf{x} \sim D(0, \sigma^2)$ (and is NOT normally distributed).

1.1 Implications of non-normality

If the error term is non-normal it implies that the OLS estimators, $b_i, i = 1, \dots, k$, will not be t-distributed. However, if the sample size is large, $n > 30$, then from a central limit theorem

$$b_j | \mathbf{x} \sim^a N(\beta_j, V(\beta_j))$$

regardless of the error term. This means that one can still undertake hypothesis testing; however, the use of F-tests is not appropriate (and one should use the asymptotic equivalent χ^2 tests).

1.2 Detecting non-normality

One can test for normality of some residuals, e_i , by testing the skewness and excess kurtosis of these residuals. Define

$$m_j = n^{-1} \sum_{i=1}^n e_i^j \text{ for } j = 1, 2, 3, 4$$

Then define skewness as:

$$S = \frac{m_3}{m_2^{3/2}}$$

and kurtosis as

$$K = \frac{m_4}{m_2^2}$$

The Jarque-Bera test of normality is then:

$$JB = \frac{n}{6} [S^2 + (K - 3)^2 / 4]$$

NOTE:

The most usual cause of non-normality is an outlier in the data set, which shows up as apparent asymmetry. This can only be detected by plotting your residuals (which you should always do, along with plotting the dependent variable as well as ALL explanatory variables). However, this is NOT non-normality, but rather an outlier problem and ought (providing there is an explanation for the outlier) to be dealt with by including an impulse dummy at the point of the outlier. In Stata the command

`sktest variable name`

tests for normality, Appendix A show some Stata output for testing normality.

A Testing for normality of errors

Source	SS	df	MS	Number of obs	=	9,526
Model	534.877205	6	89.1462008	F(6, 9519)	=	348.04
Residual	2438.16961	9,519	.256137158	Prob > F	=	0.0000
				R-squared	=	0.1799
				Adj R-squared	=	0.1794
Total	2973.04682	9,525	.312130899	Root MSE	=	.5061

lhourpay	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
edage	.0536192	.0017302	30.99	0.000	.0502276	.0570109
age	.0624868	.0026719	23.39	0.000	.0572493	.0677243
c.age#c.age	-.0006343	.0000304	-20.84	0.000	-.0006939	-.0005746
sex						
Female	-.210526	.0103917	-20.26	0.000	-.230896	-.1901561
ethnic						
Asian	-.1410976	.0235136	-6.00	0.000	-.1871893	-.095006
Other	-.1137099	.0249926	-4.55	0.000	-.1627008	-.064719
_cons	.2533564	.0625185	4.05	0.000	.1308069	.3759059

Residuals					
	Percentiles	Smallest			
1%	-4.325174	-6.222635			
5%	-.8508219	-6.024776			
10%	-.5913078	-5.560774	Obs		9,752
25%	-.3161873	-5.23019	Sum of wgt.		9,752
50%	-.0320321		Mean		-.0843224
		Largest	Std. dev.		.7953641
75%	.2937085	2.79509			
90%	.6044928	2.917032	Variance		.6326041
95%	.8120549	2.97805	Skewness		-3.127222
99%	1.29032	3.011134	Kurtosis		18.78931

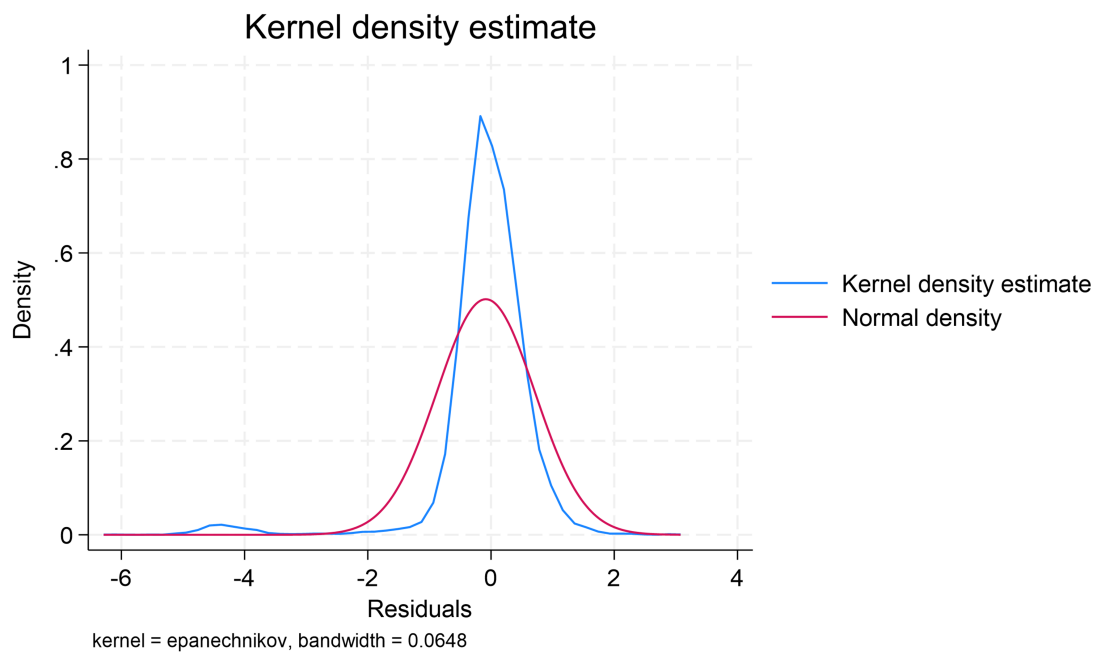
Jarque-Bera normality test: 1.2e+05 Chi(2) 0
 Jarque-Bera test for Ho: normality:

NOTE: $= \frac{9752}{6} [-3.1272^2 + (18.789 - 3)^2 / 4] = 117190.6$

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test	
				Adj chi2(2)	Prob>chi2
1hourpay_res	9,752	0.0000	0.0000	.	.

Plotting the distribution of the residuals and for a $N(0,1)$ variable as:



This plot of the distribution of the residuals (blue line) along with the distribution of a $N(0,1)$ (red line) and shows that, compared to a $N(0,1)$, there are too many observations in the very centre of the distribution and in the left tail of the distribution, with too few cases say between just above and just below median.

B Stata Commands for normality

```
/* Running a regressin and saving the residual */
regress lhourpay edage age c.age#c.age i.sex i.ethnic if(edage>=0 & edage<=95)
predict res3, resid
/* Constructing details summary statistics on the residuals */
su res3, d
/* Jarque-Bera test for the residuals */
jb res3
/* Skewness and Kurtoss test for the residuals */
sktest res3
/* Plotting the distribution of the residual vs N(0,1) */
kdensity res3, normal
```

C R Commands for normality

#In R you must install the appropriate package. Having installed the package
#you must load the library associated with that package each time you enter R.
#I used the following libraries over Term 1:

```
library(tidyverse)
library(reshape2)
library(foreign)
library(ggplot2)
library(haven)
library(readxl)
library(Hmisc)
library(ggpubr)
library(rstatix)
library(groupedstats)
library(car)
library(lmtest)
library(olsrr)
library(estimatr)
library(tseries)
library(margins)
library(ivreg)

/* Running a regressin and saving the residual */
model21 <- lm(lhourpay + edage + age + I(age^2) + sex + ethnic,
             data = subset(qlfs18_1, edage1>=0 & edage1<=95))
jarque.bera.test(resid(model21))

ggplot(qlfs18_1, aes(x=resid(model21))) +
  geom_density() +
  stat_overlay_normal_density(color = "red", linetype = "dashed")
```

References

Wooldridge, J. M. (2013). *Introduction to Econometrics*. Cengage.